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|------------------------------------------------------------------|-----------------------------------------------------|
| Program : <b>Diploma in Electronics and Computer Engineering</b> |                                                     |
| Course Code : <b>5509</b>                                        | Course Title: <b>Linear Integrated Circuits Lab</b> |
| Semester : <b>5</b>                                              | Credits: <b>1.5</b>                                 |
| Course Category: <b>Program Core</b>                             |                                                     |
| Periods per week: <b>3 (L:0, T:0, P:3)</b>                       | Periods per semester: <b>45</b>                     |

### Course Objectives:

- To equip students to acquire skills in designing and testing various practical circuits using analog integrated circuits like LM741, NE555.
- To give an outline on fixed and variable voltage regulators and digital to analog converter.

### Course Prerequisites:

| Topic                                                                       | Course code | Course name                | Semester |
|-----------------------------------------------------------------------------|-------------|----------------------------|----------|
| Working of diodes , transistors, amplifiers, multivibrators and oscillators |             | Electronic Circuits Lab    | 3        |
| Characteristics and working of Operational amplifiers                       |             | Linear Integrated Circuits | 4        |

### Course Outcomes:

On completion of the course, the student will be able to:

| CO <sub>n</sub> | Description                                                                                                           | Duration (Hours) | Cognitive Level |
|-----------------|-----------------------------------------------------------------------------------------------------------------------|------------------|-----------------|
| CO1             | Design and develop various amplifier circuits using operational amplifier                                             | 9                | Applying        |
| CO2             | Design and develop various comparators, wave shaping circuits and waveform generators using operational amplifier 741 | 15               | Applying        |
| CO3             | Develop multivibrator circuits using 555 timer IC                                                                     | 6                | Applying        |
| CO4             | Demonstrate fixed and variable power supplies and converters                                                          | 9                | Understanding   |
|                 | Lab Exam                                                                                                              | 6                |                 |

**CO-PO Mapping:**

| <b>Course Outcomes</b> | <b>PO1</b> | <b>PO2</b> | <b>PO3</b> | <b>PO4</b> | <b>PO5</b> | <b>PO6</b> | <b>PO7</b> |
|------------------------|------------|------------|------------|------------|------------|------------|------------|
| <b>CO1</b>             | 3          |            |            |            |            |            |            |
| <b>CO2</b>             | 3          | 3          | 3          |            |            |            |            |
| <b>CO3</b>             | 3          |            |            |            |            |            |            |
| <b>CO4</b>             | 2          |            |            |            |            |            |            |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

**Course Outline:**

| <b>Module Outcomes</b> | <b>Description</b>                                                                                                                                                                   | <b>Duration (Hours)</b> | <b>Cognitive Level</b> |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|------------------------|
| <b>CO1</b>             | <b>Design and develop various amplifier circuits using operational amplifier</b>                                                                                                     |                         |                        |
| M1.01                  | Setup a Voltage follower circuit using Op-Amp 741<br>1. Plot the I/O waveforms<br>2. Measure the gain<br>3. Find out the phase difference between input and output                   | 1                       | Applying               |
| M1.02                  | Design and setup an Inverting Amplifier circuit using Op-Amp 741<br>1. Plot the I/O waveforms<br>2. Measure the gain<br>3. Find out the phase difference between input and output    | 1                       | Applying               |
| M1.03                  | Design and setup a Non-Inverting Amplifier circuit using Op-Amp 741<br>1. Plot the I/O waveforms<br>2. Measure the gain<br>3. Find out the phase difference between input and output | 1                       | Applying               |
| M1.04                  | Construct a Summing amplifier circuits using Op-Amp 741 and verify the output                                                                                                        | 3                       | Applying               |
| M1.05                  | Construct Difference amplifier circuits using Op-Amp 741 and verify the output                                                                                                       | 3                       | Applying               |
| <b>CO2</b>             | <b>Design and develop various comparators, waveshaping circuits and waveform generators using operational amplifier 741</b>                                                          |                         |                        |
| M2.01                  | Setup a Zero crossing detector circuit using Op-Amp 741 and plot the I/O waveforms                                                                                                   | 1                       | Applying               |
| M2.02                  | Design a Schmitt trigger circuits using Op-Amp 741                                                                                                                                   | 2                       | Applying               |

|            |                                                                                                                                                    |     |          |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
|            | 1. Plot the I/O waveforms<br>2. Measure the LTP and UTP of the Schmitt trigger                                                                     |     |          |
| M2.03      | Design a Differentiator circuit using Op-Amp 741 and plot the pulse and frequency response                                                         | 1   | Applying |
| M2.04      | Design an Integrator circuit using Op-Amp 741 and plot the pulse and frequency response.                                                           | 1   | Applying |
| M2.05      | Design a triangular wave generator using Op-amp 741 and measure its frequency of operation                                                         | 1   | Applying |
| M2.06      | Design a monostable multivibrator using Op-amp 741<br>1. Plot the waveforms<br>2. Measure the time delay                                           | 3   | Applying |
| M2.07      | Design a symmetrical astable multivibrator using Op-Amp 741<br>1. Plot the waveforms<br>2. Find out the frequency of oscillation                   | 1.5 | Applying |
| M2.08      | Design an asymmetrical astable multivibrator using Op-Amp 741<br>1. plot the waveforms<br>2. Find out the frequency of oscillation                 | 1.5 | Applying |
| M2.09      | Design a RC phase shift oscillator using Op-Amp 741<br>1. Plot the output waveform<br>2. Measure the frequency of oscillation                      | 1.5 | Applying |
| M2.10      | Design a Wien bridge oscillator using Op-Amp 741<br>1. Plot the output waveform<br>2. Measure the frequency of oscillation                         | 1.5 | Applying |
|            | Lab Exam– I                                                                                                                                        | 3   |          |
| <b>CO3</b> | <b>Implement multivibrator circuits using 555 timer IC</b>                                                                                         |     |          |
| M3.01      | Design a symmetrical astable multivibrator using IC 555<br>1. Plot the output waveform<br>2. Measure the frequency and duty cycle of oscillation   | 1.5 | Applying |
| M3.02      | Design an asymmetrical astable multivibrator using IC 555<br>1. Plot the output waveform<br>2. Measure the frequency and duty cycle of oscillation | 1.5 | Applying |

|            |                                                                                                            |   |               |
|------------|------------------------------------------------------------------------------------------------------------|---|---------------|
| M3.03      | Design a monostable multivibrator using 555 IC<br>1. Plot the output waveform<br>2. Measure the time delay | 3 | Applying      |
| <b>CO4</b> | <b>Demonstrate fixed and variable power supplies and converters</b>                                        |   |               |
| M4.01      | Setup a +5V, 1A power supply using IC 7805                                                                 | 3 | Understanding |
| M4.02      | Setup a low voltage power supply using LM 723                                                              | 3 | Understanding |
| M4.03      | Construct a 4-bit R-2R ladder DAC and obtain the output voltage level for all input combinations           | 3 | Understanding |
|            | Lab Exam – II                                                                                              | 3 |               |

#### Text / Reference:

| T/R | Book Title/Author                                                                                                             |
|-----|-------------------------------------------------------------------------------------------------------------------------------|
| T1  | Roy D. C. and S. B. Jain, <b>Linear Integrated Circuits</b> , New Age International, 3/e, 2010                                |
| T2  | Sergio Franco, <b>Design with operational amplifiers and analog integrated circuits</b> , 3rd Edition, Tata McGraw-Hill, 2008 |
| R1  | Gayakwad R. A., <b>Op-Amps and Linear Integrated Circuits</b> , Prentice Hall, 4/e, 2010                                      |
| R2  | Botkar K. R., <b>Integrated Circuits</b> , 10/e, Khanna Publishers, 2010                                                      |
| R3  | Salivahanan S., V. S. K. Bhaaskaran, <b>Linear Integrated Circuits</b> , Tata McGraw Hill, 2008                               |
| R4  | David A. Bell, <b>Operational Amplifiers &amp; Linear ICs</b> , Oxford University Press, 2nd edition, 2010                    |

#### Online Resources:

| Sl.No | Website Link                                                                                                                                                                                          |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | <a href="http://vlab.amrita.edu/?sub=3&amp;brch=225">http://vlab.amrita.edu/?sub=3&amp;brch=225</a>                                                                                                   |
| 2     | <a href="http://he-coep.vlabs.ac.in/List%20of%20experiments.html?domain=ElectronicsandCommunications">http://he-coep.vlabs.ac.in/List%20of%20experiments.html?domain=ElectronicsandCommunications</a> |
| 3     | <a href="https://www2.mvcc.edu/users/faculty/jfiore/et262.html">https://www2.mvcc.edu/users/faculty/jfiore/et262.html</a>                                                                             |
| 4     | <a href="https://freevidelectures.com/course/2915/linear-integrated-circuits">https://freevidelectures.com/course/2915/linear-integrated-circuits</a>                                                 |

## **Student Activity**

### **Suggested Open-ended Experiments:**

Students can do open ended experiments as a group of 3-5. There is no duplication in experiments in between groups. This is mainly for the purpose of continuous internal evaluation and a score of 15 marks. Students should prepare a separate report on open ended experiment of their choice.

Example :

1. Lamp Flasher circuit(lamp dimmer circuit – where the lamp is turned ON/OFF at set frequency or time intervals) using NE 555.
2. 10 minute timer circuit using 555
3. Temperature control DC fan using 741.
4. Stereo balance indicator using 741.

### **Sample Questions to Test Outcomes**

1. Given IC 555 construct a waveshaping circuit to generate a pulse of duration  $T=1$  ms.
2. Given IC 741, generate a square wave with duty cycle 75% which oscillates at a frequency of 1kHz.